

PIN Diode IV Characteristics & Transient Analysis



Department of Electrical Engineering
Indian Institute of Technology Bombay

Background Theory

- PIN diode has a wide, un-doped intrinsic semiconductor region between 'P' and 'N' regions. The P-type and N-type regions are typically heavily doped.



- PIN diode obeys the standard diode equation for low-frequency signals. At higher frequencies, the diode looks like an almost perfect resistor.

Aim of the experiment

- To find forward voltage, reverse saturation current and ideality factor of the given PIN diode (Infineon BAR 15-1) and compare with normal PN junction Diode (1N4007).
- To find reverse recovery time of the given PIN diode at various frequencies and compare with the reverse recovery time of normal PN junction Diode.

Reverse Recovery Time

- When a diode is switched from forward to reverse bias, ideally it should switch instantly and block any current flow. But in reality, a reverse current will flow for a short period of time.

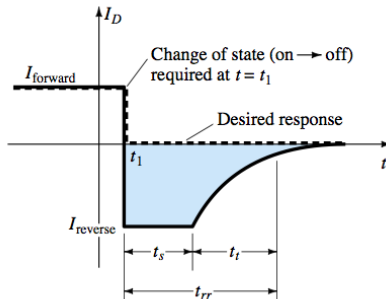
- This is caused due to stored charge in the junction, which takes time to recombine.

Reverse recovery time of the diode is the time gap between the instant at which the reverse current starts to flow through the diode to the time instant at which it reaches to 0 (or some predefined value close to 0).

- It can be a significant source of loss in switching regulations. But it can also be beneficial in some signal switching applications.

Reverse Recovery Time

The below figure illustrates what happens to the current flowing through a diode when the polarity of voltage across the diode is switched.



I_{rr} is the peak reverse current. Here, the RRT is measured till when the reverse current reaches 25% of I_{rr} .

Components required

- PIN diode Infineon BAR 15-1
- Resistors - 500Ω ($\times 2$), 50Ω , $1k\Omega$
- Potentiometer - $1k\Omega$
- Capacitors - $100nF$ ($\times 2$)
- Breadboard, connecting wires
- Multi-meters, variable power supply, signal generator and oscilloscope

Experiment-Part 1

- 1 Make the connections as per the circuit diagram. Notice the role of the $1k\Omega$ pot, that can be used to vary the voltage to be applied to the diode.
- 2 Vary V_D in suitably small steps (from 0 V to 1 V only), and measure and tabulate I_D and V_D for each step.

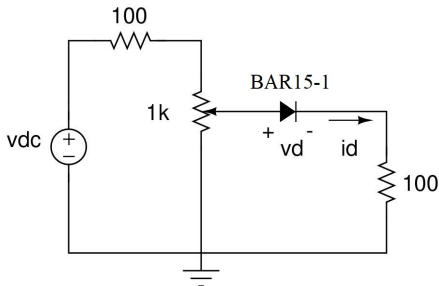


Figure: Circuit to measure I/V characteristics of a semiconductor diode

You are advised to simultaneously plot I-V in the lab, to quickly identify if the measurement is as expected.

Experiment-Part 2

Apply square wave of different frequencies (10 *KHz*, 100 *kHz*, 1 *MHz*, 3 *MHz*) to the given circuit (for both PIN diode and PN diode) and note down the reverse recovery times.

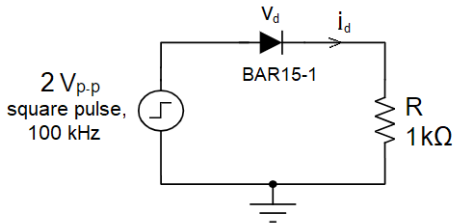


Figure: Circuit to measure reverse recovery time

Which diode has the potential of passing major portion of the input signal to the output at 3 *MHz*?

BAR15-1 Device

The Infineon BAR15-1 PIN Diode has a common cathode structure.

